

Mohan Shankar G

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Profile

Full Stack Developer with 3+ years of experience in developing scalable web applications, AI-powered computer vision systems, and embedded solutions. Proficient in React.js, Node.js, Express.js, MongoDB, PostgreSQL, Python, and RESTful APIs. Experienced in government, enterprise, and AI-based projects with strong problem-solving and collaboration skills.

Education

Bachelor of Technology in Computer Science and Engineering — Andhra University, Visakhapatnam. Passed Out in 2019

Intermediate (MPC) — Narayana Junior College, Nidadavole. Passed Out in 2015

Experience

Full Stack Developer — Satra Service and Solutions Pvt. Ltd

- Working on the GujMarg Project, a Gujarat Government initiative for road safety and complaint management.
- Developed complaint management modules, automated officer assignment, and escalation tracking up to the CMO.
- Contributed to both backend development using Node.js and Express.js and front-end development using React.js.
- Worked on AI/ML-based Vehicle Detection systems using the BoT-SORT algorithm for object tracking.
- Performed data annotation using LabelImg and Label Studio.
- Built vehicle detection and tracking pipelines using Python and computer vision techniques.
- Contributed to traffic analysis systems for identifying high-density roads across India.
- Maintained codebase and version control using GitHub.

Backend Developer — Synozon Technology

- Developed full-stack CRM modules for customer interaction tracking and relationship management.
- Created backend APIs using Node.js and Express.js.
- Integrated SAP ABAP for B2B ticket synchronization.
- Implemented responsive React.js user interfaces for seamless CRM operations.
- Developed order management and customer profile modules.

Embedded Developer / Jr. Software Engineer — Pantech Solutions R&D

- Worked on EV vehicle prototype design integrating embedded systems and AI.
- Developed assistive robotics projects including Blind Man Wheelchair and Auto Robo Assistant.
- Worked with Embedded C, Python, C++, AI/ML, and Data Science technologies.
- Collaborated on hardware integration, sensor systems, and automation.

Projects

GujMarg — Gujarat Government Road Safety Portal

- Built a complaint management system with automated officer assignment and hierarchical escalation up to the CMO.
- Developed REST APIs using Node.js and Express.js.
- Implemented complaint sorting, filtering, and role-based access control.
- Created Cron Jobs for daily officer ranking and automatic archival of records.
- Developed React.js dashboards, Web Audit, and Citizen Audit views.
- Designed responsive interfaces with tables, filters, modals, and pagination.

CRM — Customer Relationship Management System

- Developed a full-stack CRM platform using React.js, Node.js, and MongoDB.
- Created REST APIs for orders, alerts, authentication, and customer profiles.
- Integrated SAP ABAP for enterprise-level order tracking and business process automation.
- Implemented secure authentication and notification modules.
- Designed backend architecture for efficient data handling and synchronization.
- Maintained customer profile and order management modules.

Nirupaa — Full-Stack Fashion E-Commerce Platform

- Developing and maintaining a scalable fashion e-commerce platform for men's and women's clothing.
- Built responsive UI with product browsing, filtering, cart, and checkout functionalities.
- Implemented secure JWT-based authentication and admin dashboard.
- Developed REST APIs using Node.js, Express.js, and MongoDB.
- Integrated Cloudinary, Multer, and Sharp for image management.
- Deployed frontend and backend services on Vercel.

Vehicle Detection using AI & ML

- Developed an AI-based vehicle detection and tracking system using the BoT-SORT algorithm.
- Performed data annotation using LabelImg and Label Studio.
- Built object detection and tracking pipelines using Python and OpenCV.
- Implemented traffic analysis for identifying high-density roads across India.
- Designed the system to support large-scale traffic survey applications.
- Maintained source code and collaboration using GitHub.

Road Furniture Asset Detection System

- Developed an AI-based road asset detection and classification system using YOLOv11.
- Built backend services using Node.js, Express.js, PostgreSQL, Prisma ORM, and Redis.
- Implemented batch inference pipelines for scalable processing.
- Implemented Redis-based queue processing and batch inference pipelines.
- Generated reports and annotated outputs for large-scale road surveys.
- Designed the system for infrastructure analysis and asset inventory management.
- Utilized GitHub for source control and collaborative development.

Blind Man Wheelchair

- Designed an AI-enabled wheelchair to assist visually impaired users through obstacle detection and navigation.
- Integrated ultrasonic and IR sensors for real-time environment monitoring.
- Implemented motor control and automatic stopping mechanisms.
- Developed Embedded C and Python programs for hardware control.
- Performed prototype testing and hardware debugging.

Auto Robo Assistant

- Developed an autonomous robot using embedded sensors and AI techniques.
- Implemented obstacle avoidance and motion control algorithms.
- Designed safety mechanisms including auto-stop and collision prevention.
- Developed Embedded C and Python programs for hardware management.
- Optimized robot movement and stability through testing and debugging.

Skills

- **Programming Languages:** JavaScript, Python, C, C++, Embedded C
- **Frontend Technologies:** React.js, Angular.js, HTML5, CSS3, Material UI, Axios, Vite, Responsive Web Design, Front-End Development
- **Backend Technologies:** Node.js, Express.js, RESTful APIs, JWT Authentication, Serverless Functions, Web Backend Development, Cron Jobs, Role-Based Access Control (RBAC)
- **Databases:** MongoDB, PostgreSQL, MySQL, Redis
- **AI/ML & Computer Vision:** Python, YOLO, YOLOv11, BoT-SORT, OpenCV, Object Detection, Object Tracking, Vehicle Tracking, Computer Vision, Data Annotation, LabelImg, Label Studio
- **Cloud & Deployment:** Vercel, Cloudinary, Render
- **Tools & Technologies:** Git, GitHub, Postman, API Testing, Prisma ORM, Multer, Sharp, SAP ABAP Integration
- **Embedded Systems:** Arduino, Sensor Integration, Robotics, Motor Control, Embedded System Design
- **Software Development Practices:** API Development, API Integration, Version Control, Debugging, Authentication, Problem Solving
- **Soft Skills:** Team Collaboration, Communication, Leadership, Time Management, Working Under Pressure, Quick Learning

Achievements

- Awarded Best Trainee Certificate from Karpagam Engineering College, Coimbatore.
- Participated in Robotics programs and project presentations at Karpagam Engineering College.
- Won 1st place in NBA(Nidadavole Basketball Assoc), (Nidadavole Cricket Assoc).

Interests

Development, Coding, Learning New Technologies, Artificial Intelligence, Computer Vision, Movies, Basketball, Cricket, Hockey, Chess